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MILITARY SPECIFICATION

NIGHT VISION SIGHT
CREW SERVED WEAPONS
AN/TVS-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the night vision sight for crew served weapons. (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, Flat and Seals
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-636	- Boxes, Shipping Fiberboard
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triplewall
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	- Strapping, Nonmetallic, (And Connectors)
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (CDR, USA Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, NJ 07703) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

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MIL-P-116	- Preservation - Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-P-19644	- Plastic Molding Material (Polystyrene Foam Expanded Bead)
MIL-P-19834	- Plates, Indentification, Metal Foil, Adhesive Backed
MIL-P-26514	- Polyurethane Foam, Rigid or Flexible for Packaging

STANDARDS

FEDERAL

FED-STD-356	- Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-202	- Test Methods of Electronic and Electrical Component Parts
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-705	- Generator Sets, Engine-Driven, Methods of Tests and Instruction
MIL-STD-810	- Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-645660	- Night Vision Sight Crew Served Weapons Assembly

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Description. The night vision sight shall consist of a viewer, manual, and a carrying case. The sight shall consist of an in-line and *right angle

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eyepiece, objective lens assembly, boresight assembly, high-voltage oscillator, reticle and battery.

*The right angle eyepiece shall be considered a component of the viewer only when specified in the contract or order. (see 6.2).

3.2 Construction. The assembly shall be in accordance with SC-DL-645660 and all provisions of this specification.

3.3 First article. The supplier shall furnish and test assemblies as specified in the contract or purchase order (see 6.2).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Operational and Environmental.

3.5.1 Modulation transfer function-objective lens assembly. The objective assembly modulation transfer function (see 6.3.4) measured as described in figure 1 shall equal or exceed the curve shown in figure 2.

3.5.2. Modulation transfer function-eye-piece lens assembly. The eyepiece assembly modulation transfer function measured as described by figure 1 shall equal or exceed the curve shown on figure 3 when the eyepiece is at zero diopter, ±0.5 diopter and the eyepiece lens is away from the photodetector and facing the collimator.

3.5.3 Peripheral Magnification. The magnification at 2.25° from the center of the field of view shall not be less than 7.80 x nor greater than 8.30 x for the in-line eyepiece and not less than 7.16 x nor greater than 7.67 x for the right-angle eyepiece.

3.5.4 Field of view. The field of view shall be not less than 100 milliradians.

3.5.5 Stray Light Luminance. Stray light luminance contributions through the system over the image of a disk illuminated at 5×10^{-3} foot-lambert subtending a solid angle of 6.5°, plus or minus 0.5° with a white surrounding surface illuminated at 5×10^{-3} foot-lamberts subtending a solid angle of 120° minimum shall not exceed 3 percent of the luminance of the illuminated disk measured through the system without the white surrounding surface illuminated (see figure 4).

3.5.6 Focus Adjustment. Focus adjustment shall insure:

a. Viewing of sharp clear images of objects located at distances from 50 meters to infinity. Torque shall not exceed 7.0 inch-pounds.

b. Eyepiece focus of plus or minus 4 diopters by applying not more than 7.0 pound-inches or torque with an indexing error not exceeding plus or minus 0.5 diopter.

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3.5.7 Boresight Adjustment. The viewer shall be adjustable over not less than 4° in both elevation and azimuth. Adjustment in azimuth shall not change elevation, nor shall adjustment of elevation change the azimuth. The reticle patterns shall be in clear sharp focus. There shall be no parallax between the reticle and image.

3.5.8 Click Movement. Adjustment of boresight shall be indexed by means of audible clicks. Boresight knob torque shall be 8.0 inch-pounds maximum, and 3.0 inch-pounds minimum.

3.5.9 Electromagnetic Interference. The electromagnetic compatibility of the viewer shall be in accordance with MIL-STD-461, Table A-I-MIL-E-55301, equipment class-Tactical-Non-CE.

3.5.10 Reticle Intensity. The reticle intensity shall be adjustable such that the reticle pattern shall be clearly visible coincidently with with background varying between 1×10^{-2} foot-lambert and 1×10^{-5} foot-lambert brightness.

3.5.11 Shock. The viewer shall not be damaged (see 6.3.1) when subjected to installation and use on the specified weapons under each of the following conditions:

a. The firing of 500 roungs from the .50 caliber machine gun in the ground-mounted mode of operation.

b. When the right angle eyepiece is specified, the firing of 50 rounds from the major bore of the 106 mm recoiless rifle in the ground-mounted mode of operation.

3.5.12 Ruggedness. The viewer in its carrying case shall not be damaged after vibration for 30 minutes in accordance with MIL-STD-705, method 470.1a, except that test method 614.1 shall not apply.

3.5.13 Vibration. The viewer shall not be damaged after vibration at a frequency of 30 hertz (Hz) at an amplitude of not less than 0.0625 inch (total displacement of 0.1250 inch) for periods of 10 minutes each along each of the following axes:

a. Parallel to the optical axis.

b. Perpendicular to the optical axis in the vertical plane.

3.5.14 Environmental. The viewer shall not be damaged (see 6.3.2) by storage, operation, thermal shock or humidity and shall meet specified operational requirements under the following conditions (see figures 5 and 6):

a. Storage: At plus 68°C for 4 hours and at minus 54°C for 12 hours with humidity controlled at 90 to 98 percent at and above room temperature (see 6.3.3) and uncontrolled humidity below room temperature.

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b. Operation:	At temperatures from plus 52°C to minus 54°C at maximum humidities possible throughout the temperature range and atmospheric pressures from 30 to 5.5 inches of mercury.
c. Thermal Shock:	From room temperature to plus 68°C; from plus 52°C to room temperature; from room temperature to minus 54°C and from minus 54°C to room temperature.
d. Humidity:	From room temperature to plus 52°C, and from plus 52°C to room temperature at 90 to 98 percent relative humidity.

3.5.15 Immersion. The viewer shall be watertight, shall operate as specified herein, and shall not be damaged after immersion to a depth of 3 feet, for 30 minutes in fresh water.

3.5.16 Salt fog . The viewer shall operate as specified herein and shall not be damaged after having been tested in accordance with method 509 of MIL-STD-810.

3.6 Government Loaned Property. The following property in the quantities indicated will be loaned by the Government (see 6.4).

ITEM NO.	DESCRIPTION	STOCK NO. (FSN)	QUANTITY
1.	Machine Gun, Cal. .50 HB, M2	1005-726-5636	2
2.	M2 Bracket	5855-911-0703	2
3.	Mount, Tripod, M3	1005-322-9716	2
4.	Cartridge, Cal. .50, Link Ball, M2	1305(A555) Machine-Gun Grade	As Required
5.*	Gun, 106 mm, M40-A1 with .50 Cal. spotting Rifle MBC	1015-348-4925	2
6.*	M40 Bracket	5855-911-0704	2
7.*	High Explosive Plasticized Tracer Round (HEP-T) for 106 mm Rifle M40	1315(AC651)	As Required
8.	Image Intensifier Assembly	5855-051-2792	10

*Item Nos. 5, 6, and 7 are applicable only when the right angle eyepiece is specified.

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3.7 Identification Marking. Each viewer shall be marked for identification in accordance with MIL-P-19834 and as shown on the applicable drawings.

3.8 Items to be Furnished with Viewer. No shipment of viewers shall be made unless such repair parts, maintenance tools, operating tools, technical publications, and accessories which are to be furnished with each viewer are included with the shipment or unless approval for shipment of the viewers without such items has been received from the contracting officer.

3.9 Workmanship. All parts, components, and assemblies of the viewer including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, optics and electrical wiring and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous materials. Joints and seams shall be tight; all edges shall be smoothly finished. The interior of the viewer shall be free of all foreign matter.

4. - QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. Inspection shall be classified as follows:

a. First Article Inspection (see 4.3).
b. Quality Conformance Inspection (see 4.4).
c. Inspection of Preparation for Delviery (see 4.6).

4.3 First Article Inspection.

4.3.1 Inspection. Each first article viewer shall be inspected for defects, in accordance with Table I. Presence of one or more defects shall be cause for rejection.

TABLE I. Inspection.

Defects	Requirement Paragraph
MAJOR	
101. Boresight adjustment knob binds.	3.5.7.1

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TABLE I. Inspection. (Cont'd)

Defects		Requirement Paragraph
102.	Objective binds while focusing.	3.5.6
103.	Eyepiece diopter adjustment binds.	3.5.6
104.	External hardware on viewer loose.	3.2
105.	Intensifier does not energize when viewer is turned ON.	3.4
106.	Reticle fails to illuminate when switch is in the "reticle-on" position.	3.2
107.	Finish on viewer and carrying case does not conform to applicable drawings.	3.2
108.	Identification marking not as specified	3.7

4.3.2 Tests. Following successful completion of the examination specified in 4.3.1 each first article viewer shall be subjected to all the tests in Table II by the contractor. First article failure of any test shall be cause for rejection of the viewer. The first 6 tests listed in Table II shall be conducted first and in the order listed. The remaining test may be performed in any order except that contrast transfer tests shall be conducted last.

TABLE II. First Article Inspection.

Inspection	Requirement paragraph	Test paragraph
Modulation Transfer Objective	3.5.1	4.5.1
Modulation Transfer Eyepiece	3.5.2	4.5.2
Ruggedness	3.5.11	4.5.12
Shock	3.5.10	4.5.11
Vibration	3.5.12	4.5.13
Environmental	3.5.13	4.5.14
Immersion	3.5.14	4.5.15
Peripheral Magnification	3.5.3	4.5.3
Field of View	3.5.4	4.5.4
Focus Adjustment	3.5.6	4.5.6
Boresight Adjustment	3.5.7	4.5.7
Click Movement	3.5.8	4.5.8
Stray Light Luminance	3.5.5	4.5.5

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TABLE II. First article inspection (Cont'd)

Inspection	Requirement paragraph	Test paragraph
Electromagnetic Compatibility	3.5.9	4.5.9
Salt Fog	3.5.15	4.5.16
Reticle Intensity	3.5.9	4.5.10
Modulation Transfer-Objective	3.5.1	4.5.1
Modulation Transfer-Eyepiece	3.5.2	4.5.2

4.4 Quality Conformance Inspection.

4.4.1 Examination. Each viewer shall be examined as specified in 4.3.1 Presence of one or more defects shall be cause for rejection.

4.4.2 Tests.

4.4.2.1 Group A. Each viewer shall be tested in accordance with Table III. Failure of any test shall be cause for rejection.

TABLE III. GROUP A Inspection.

Inspection	Requirement paragraph	Test paragraph
Contrast Transfer-Objective	3.5.1	4.5.1
Contrast Transfer-Eyepiece	3.5.2	4.5.2

4.4.2.2 Group B. Group B tests shall be conducted on viewers selected from units which have passed the examination and tests in 4.4.1 and 4.4.2. Sampling shall be in accordance with MIL-STD-105, AQL 2.5 percent defective, inspection level S3. Tests shall be conducted in accordance with Table IV.

TABLE IV. GROUP B Inspection.

Inspection	Requirement paragraph	Test paragraph
Immersion	3.5.14	4.5.15
Vibration	3.5.12	4.5.13
Contrast Transfer-Objective	3.5.1	4.5.1
Contrast Transfer-Eyepiece	3.5.2	4.5.2

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4.4.2.3 Group C. This inspection shall consist of the tests specified in Table V and shall be performed on units that have been submitted to and which have met Group A inspection. Units shall be selected in accordance with 4.4.2.3.1.

4.4.2.3.1 Sampling for Inspection. Three viewers shall be selected by the Government for all inspections in Table V whenever the monthly quantity specified in the contract starts quality conformance testing or every 30 days whichever occurs first. The three units are not associated with inspection lots as prescribed in MIL-STD-105. The only selection prerequisite is that the three units conform to normal production processing. The first three Group C units shall begin test at the same time the first quality conformance lot (see 4.4.2.2) begins Group B testing.

TABLE V. GROUP C Inspection.

Inspection	Requirement paragraph	Test paragraph
Shock	3.4	4.5.11
Environmental	3.7	4.5.14
Peripheral Magnification	3.5.3	4.5.3
Field of View	3.5.4	4.5.4
Stray Light	3.5.5	4.5.5
Focus Adjustment	3.5.6	4.5.6
Boresight Adjustment	3.5.7	4.5.7
Click Movement	3.5.7.2	4.5.8
Salt Fog	3.5.15	4.5.16
Reticle Intensity	3.5.9	4.5.10
Modulation Transfer-Objective	3.5.1	4.5.1
Modulation Transfer-Eyepiece	3.5.2	4.5.2

4.4.2.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2).

4.4.3 Reinspection of Group B and Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5 Test methods. All tests shall be conducted at room temperature (see 6.3.3) unless otherwise specified in the individual test procedure. Test facilities, chambers and apparatus used in conducting tests specified shall meet the requirements of MIL-STD-810.

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4.5.1 Modulation transfer-objective lens assembly . The modulation transfer of the objective lens assembly shall be measured using the equipment prescribed in figure 1. Failure to equal or exceed the continuous curve of figure 2 shall constitute failure of the test.

4.5.2 Modulation transfer-eyepiece lens assembly. The modulation transfer of the eyepiece lens assembly shall be measured with the equipment prescribed in figure 1. The eyepiece lens shall be positioned away from the photo-detector and facing the collimator. The diopter adjustment shall be set at zero, $\pm$0.5 diopter. Failure to equal or exceed the continuous curve of figure 3 shall constitute failure of the test.

4.5.3 Peripheral Magnification. The apparatus shown in Figure 7 shall be used to measure the peripheral magnification. With the viewer in place on-axis as shown in figure 7 view the target pattern through the viewer with a telemicroscope having a focal length equal to the eyepiece focal length. The focal length of the collimator lens shall be 238.5 mm and the diameter of the test pattern shall be 20.0 mm. The telemicroscope shall have two concentric circles located at the focal plane or the objective lens with apparent diameters of 22.20 and 23.60 mm for the in-line eyepiece and diameters of 20.20 and 21.50 mm for the right-angle eyepiece. Failure of the test target to appear between the two concentric circles, in the telemicroscope, shall constitute failure of this test.

4.5.4 Field of View. The test instrument shown in figure 8 shall be used to project a spot of light or point source onto the objective lens of the viewer. The viewer shall then be pivoted either to the left or right until the point source seen through the viewer is at the edge of the field of view. The viewer shall then be pivoted in the opposite direction until the point source is at the opposite edge of the field of view. The total angle through which the viewer was pivoted shall be measured and the values recorded. A field of view less than 100 angular mils shall constitute failure of this test.

4.5.5 Stray Light Luminance. The apparatus shown in figure 4 shall be used to measure the stray light luminance. The 2870° K tungsten light source (No. 1) employed in illuminating the white surrounding surface shall be positioned so as to provide a nearly uniform brightness of 5×10^{-3} footlambert over the surface. The 2870° K tungsten light source (No. 2) employed in illuminating the disk shall be positioned so as to provide a uniform brightness of 5×10^{-3} footlambert over the surface. The eyecorrected photodetector (see 6.3.5) shall have an aperture of 7 mm. Adjust the objective lens of the viewer for infinity focus. Compare the viewer light output with the light source (No. 1) OFF and light source (No. 1) ON. The measurements shall be made with light source (No. 2) illuminating the surface of the disk. Any change of more than 3.0 percent in light output in the measurements with light source (No. 1) OFF and the light source (No. 1) ON shall constitute failure of this test.

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NOTE: Care shall be taken in arranging the illuminating source (light source No. 1) so that no light falls on the surfaces illuminated by light source No. 2.

4.5.6 Focus Adjustment.

4.5.6.1 Objective Lens. The viewer shall be focused on a collimated image by adjusting the objective focus and adjusting the eyepiece diopter ring. The viewer shall also be focused on an object at a distance not greater than 50 meters by adjusting the objective focus and adjusting the eyepiece diopter ring. Inability of the objective lens to sharply focus a collimated image simulating infinity and an object at not more than 50 meters from the viewer shall constitute failure of this test.

4.5.6.2 Eyepiece Lens. The viewer objective lens shall be focused on an object by viewing a real or collimated image. Adjust the eyepiece to an extreme diopter setting as indicated on the eyepiece. Position a diopter tester against the eyepiece lens and adjust until the object viewed through the viewer and diopter tester com bination is in clear focus. Repeat the above procedure after adjusting the eyepiece lens for the other extreme diopter setting. Failure of the eyepiece readings to coincide with the diopter tester readings within plus or minus 0.50 diopter or failure of the eyepiece to be adjustable from plus 4, to minus 4 diopters shall constitute failure of this test.

4.5.7 Boresight Adjustment. Use the test equipment shown in figure 9 with the target consisting of a circle and crosshairs subtending 4° of the field of view. The target shall be concentric with the optical axis of the telescope. Adjust the center of the reticle to the extremities of the 4° circle. Failure of the boresight to obtain this adjustment shall constitute failure of this test.

4.5.8 Click Movement. The test fixture described in figure 7 shall be used to measure the boresight click movement. Measurements shall be taken at both extremes and in the center of the adjustment for both elevation and azimuth. Determine the center of the azimuth adjustment. Turn the azimuth adjustment to a point between 40 and 50 clicks from the center. Aline the viewer so that the reticle pattern is centered on the main crosshairs of the target. Turn the azimuth adjustment one click in either direction and observe the position of the crosshairs relative to the two lines A & B that are applicable. Turn the azimuth adjustment two clicks in the opposite direction and again observe the position of the crosshairs relative to lines A & B this time 180° from the previous A & B lines. Repeat the test for a point within 5 clicks of the center of adjustment and for a point between 40 and 50 clicks on the opposite side of the center. Repeat the six position procedure for the elevation adjustment. Failure of the reticle pattern to fall between the two lines A & B for any one of the twelve observations on the target or failure to meet the torque requirements of 3.5.8 shall constitute failure of the test.

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4.5.9 Electromagnetic Interference. The first article viewer, equipped for the reduction of radio interference shall be tested as specified in 3.5.9. The supplier shall furnish the contracting officer the reports requried by MIL-STD-461, except that the time interval for submission of reports shall be as specified (see 6.2). Upon approval of the reports by the contracting officer and provided all other requirements of the specification are met, the first article viewer shall be used as a model for all production viewers.

4.5.10 Reticle Intensity. Remove the collimator lens and resolution pattern from the apparatus shown in figure 9. Place a diffusing screen between the light source and the viewer of such transmission that a brightness between 1×10^{-4} and 5×10^{-5} footlambert is presented to the objective lens of the viewer. Turn ON the reticle and adjust the intensity until the reticle pattern dims to invisibility. Replace the diffusing screen with one of such transmission that a scene brightness of between 1×10^{-2} and 5×10^{-2} footl ,mbers is presented to the objective lens of the viewer. Turn ON the reticle and adjust the intensity until the concidental presence of the reticle pattern and the scene is obtained. Failure of the reticle pattern to dim to invisibility or to obtain coincidental presence of the reticle pattern and the scene shall constitute failure of this test.

4.5.11 Shock. This test may be conducted during daylight hours. The number of rounds to be fired shall be as follows:

Inspection	.50 Caliber Machine Gun
a. First Article Inspection	500 rounds
b. For Quality Conformance Testing	300 rounds

4.5.11.1 Machine Gun. The viewer shall be mounted on the .50 caliber machine gun by means of a weapons adapter bracket and shall be sighted in on target with the weapon in the ground-mounted mode of operation. The sight shall be boresighted in the manner prescribed in the applicable technical manual. All hardware and connectors shall be checked to insure that no parts are loose. The weapon shall then be fired in 5-to-10 round bursts until 100 rounds have been fired. During and after the firing, observe and record any internal flickering, or deterioration in quality of the image presented by the viewer; or any loosening or failure of any hardware and adjustments. After 100 rounds have been fired, the position of the reticle in the viewer with respect to the open sight of the weapon shall be checked and compared with the relative positioning prior to firing the 100 rounds. Any change in boresight, reticle shift, reticle rotation, or other evidence of damage shall be recorded. The 100 rounds firing and examination cycle shall be repeated as often as required to accumulate the total number of rounds specified to be fired. No adjustments, tightening of parts, or repairs of any kind shall be permitted after the initial boresighting is completed and the test firing cycles have begun without recording of same. Evidence of damage or loss of alinement with the weapon shall constitute failure of this test.

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4.5.11.2 Recoiless Rifle. This test shall be conducted only when the right angle eyepiece is specified. The viewer shall be mounted on the 106 mm recoiless rifle by means of a weapon adapter bracket and shall be sighted in on target with the weapon in the ground-mounted mode of operation. The sight shall be boresighted in the manner prescribed in the applicable technical manual. All hardware and connectors shall be checked to insure that no parts are loose. The weapon shall then be fired for 10 consecutive rounds. During and after firing, observe and record any internal flashing, flickering, or deterioration in quality of the image presented by the viewer, or any loosening or failure of any hardware and adjustments. After the 10 rounds have been fired, the position of the reticle in the viewer with respect to the open sight of the weapon shall be checked and compared with the relative positioning prior to firing the 10 rounds. Any change in boresight, reticle shift, reticle rotation, or any other evidence of damage shall be recorded. The 10 round firing and examination cycle shall be repeated as often as required to accumulate the total number of rounds specified to be fired. No adjustments, tightening of parts, or repairs of any kind shall be permitted after the initial boresighting is completed and the test firing cycles have begun without recording of same. The test on the 106 mm recoilless rifle shall be conducted with the right angle eyepiece only. Evidence of damage or loss of alinement with the weapon shall constitute failure of this test.

4.5.12 Ruggedness. The ruggedness test shall be performed with all required parts present in the carrying case. The case shall be placed on a vibration table without tiedown of any kind and vibrated for 1 hour each on both the top and bottom of the case in accordance with MIL-STD-705, Method 740.1a with only the vibration part applying to this test. Evidence of damage shall constitute failure of this test.

4.5.13 Vibration. Mount the viewer rigidly to the vibration table. Subject the viewer to a simple harmonic motion parallel to the optical axis and perpendicular to the optical axis in the vertical plane at an amplitude of 0.0625 inch (0.1250 inch total excursion) at frequency of 30 Hz for 5 minutes in each position. Examine the viewer. Evidence of damage shall constitute failure of this test.

4.5.14 Environmental.

4.5.14.1 Humidity. Place the viewer in a chamber at room temperature and a relative humidity of 90 to 98 percent and soak for 4 hours. Raise the temperature of the chamber to 52°C + 2°C in not more than 2 hours, maintain the relative humidity of 90 to 98 percent and soak for 4 hours. Lower the temperature of the chamber to room temperature in not more than 2 hours. This constitutes one complete cycle as shown in figure 6. Perform the above cycle six times. Turn the viewer ON and observe for image intensifier operation for the last 30 minutes of the room temperature portion of the 4th and 6th cycles. Failure to operate or evidence of damage during or immediately following the above six cycles shall constitute failure of this test.

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4.5.14.2 Thermal Shock, Operation, High and Low Temperature Storage. Place the viewer in a chamber which has an internal temperature of 68°C and soak for 4 hours. Lower the temperature in the chamber to 54°C in not more than 2 hours and soak for 4 hours. Turn the viewer ON and observe for image intensifier operation during the first 20 minutes of this 4 hour period. Remove the viewer from the chamber and soak for 2 hours at room temperature. Place the viewer in a chamber which has an internal temperature of minus 54°C. Soak the viewer at this temperature for 12 hours. Turn the viewer ON and observe for image intensifier operation for the last 30 minutes of the twelfth hour. It is permissible to replace the battery in the unit at this time, with a battery which has been at room temperature in order to operate this unit. Remove the viewer from the chamber and soak for 4 hours at room temperature. Turn the viewer ON and observe for image intensifier operation for the last 10 minutes of this period. This constitutes one complete 28 hour cycle as shown in figure 5. Repeat the above cycle. Failure to operate or evidence of damage during or immediatly following the above two cycles shall constitute failure of this test.

4.5.14.3 Barometric Pressure Reduced. Test the viewer in accordance with Method 105-C of MIL-STD-202 at an altitude of 40,000 feet (5.5 inches of mercury pressure) for 12 hours mounted horizontally. At the end of this period increase the barometric pressure to the ambient, remove the viewer from the chamber and operate it. Failure to operate as specified in 3.5.14 or evidence of any damage shall constitute failure of this test.

4.5.15 Immersion. Immerse the viewer to a minimum depth of 3 feet in fresh water for a period of 30 minutes. Remove the viewer from the water, dry all external surfaces disassemble and inspect for entrance of water or moisture. Any evidence of water or moisture within the viewer shall constitute failure of this test.

4.5.16 Salt Fog. Test the viewer in accordance with Method 509 of MIL-STD-810 using a 20 percent salt solution.

4.6 Inspection of Packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance to requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Night Vision Sight, Crew Served Weapons AN/TVS-2 shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. Night Vision Sight, Crew Served Weapons AN/TVS-2 shall be dried in accordance with applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method 1C-1.

5.1.1.4.2 Night Vision Sight, Crew Served Weapons AN/TVS-2. Each night vision sight, shall be individually unit packed method III as follows: place each component within the spaces provided in the case. Close the cover and secure the fasteners. Cushion the case on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed and fabricated to protecr all projections ans absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming the MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned case with a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservative application and unit packing. Each night Vision sight, shall be cleaned, dried, preserved and unit packed as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A . A quantity of Night Vision Sight, Crew Served Weapons AN/TVS-2, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming t PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shalll conform to QQ-S-781, type 1, finish A.

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5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.2 Level B. A quantity of Night Vision Sight, Crew Served Weapons AN/TVS-2, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming the PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets, conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Night Vision Sight, Crew Served Weapons AN/TVS-2, unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and esterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended Use. The viewer is intended for use as a night observation device. In addition to being used as a general-purpose night vision surveillance telescope, it can be used as a counter measure aid in the night detection of enemy use of active systems.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether or not the right angle eyepiece is specified.

c. Technical publications required (see 3.8).

d. Repair parts and maintenance tools required (see 3.8).

e. Government loaned property (see 3.6).

f. Schedule for first article testing (see 3.3).

g. Technical activity responsible for Group C failures (see 4.4.2.3.2).

h. Level of packaging, packing and if consolidation is required (see 5.1 and 5.2).

i. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material spec-ifications or preparing activity for recommended disposability methods.

6.3 Definitions. The following definitions shall apply throughout this specification.

6.3.1 Shock and Vibration Damage. Includes any loosening of parts, change of sighting adjustment, failure or breakage of any hardware, connection or com ponent, any change in quality of the image, any evidence of intermittent electrical contact, any change in boresight; any shift in alinement.

6.3.2 Operation and Storage Damage. Includes breakages, loosening, failure, deterioration, corrosion of any finish, hardware, optical coating, connection or component; condensation of moisture within the telescope and high voltage breakdown; failure to operate in accordance with provisions of this specification.

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6.3.3 Room Temperature. Room temperature is defined as 23°C, ± 3°C.

6.3.4 Modulation transfer function (MTF). A measure of the performance of a lens assembly as an image-producting device. In effect it is the frequency response of the assembly obtained by measuring the degradation of a slit image, as correlated to a variable frequency sine-wave pattern.

6.3.5 Eye-corrected Photo-detector. For the purpose of this specification an eyecorrected photodetector is a photodetector or brightness meter that measures light in the particular measurement concerned within plus or minus 5 percent of that which would be obtained if the measurement were made with a device which was perfectly corrected to the Standard International Commission on Illumination (C.I.E.) curve for photopic of the human eye.

6.4 Government Loaned Property. The contracting officer will arrange the delivery of Government loaned property specified in 3.6. The contracting officer should also arrange for the return of any unexpended rounds.

CUSTODIAN:
Army - EL
Navy - EC
Air Force - 99

PREPARING ACTIVITY:
Army - EL

Project No. 5855-0017

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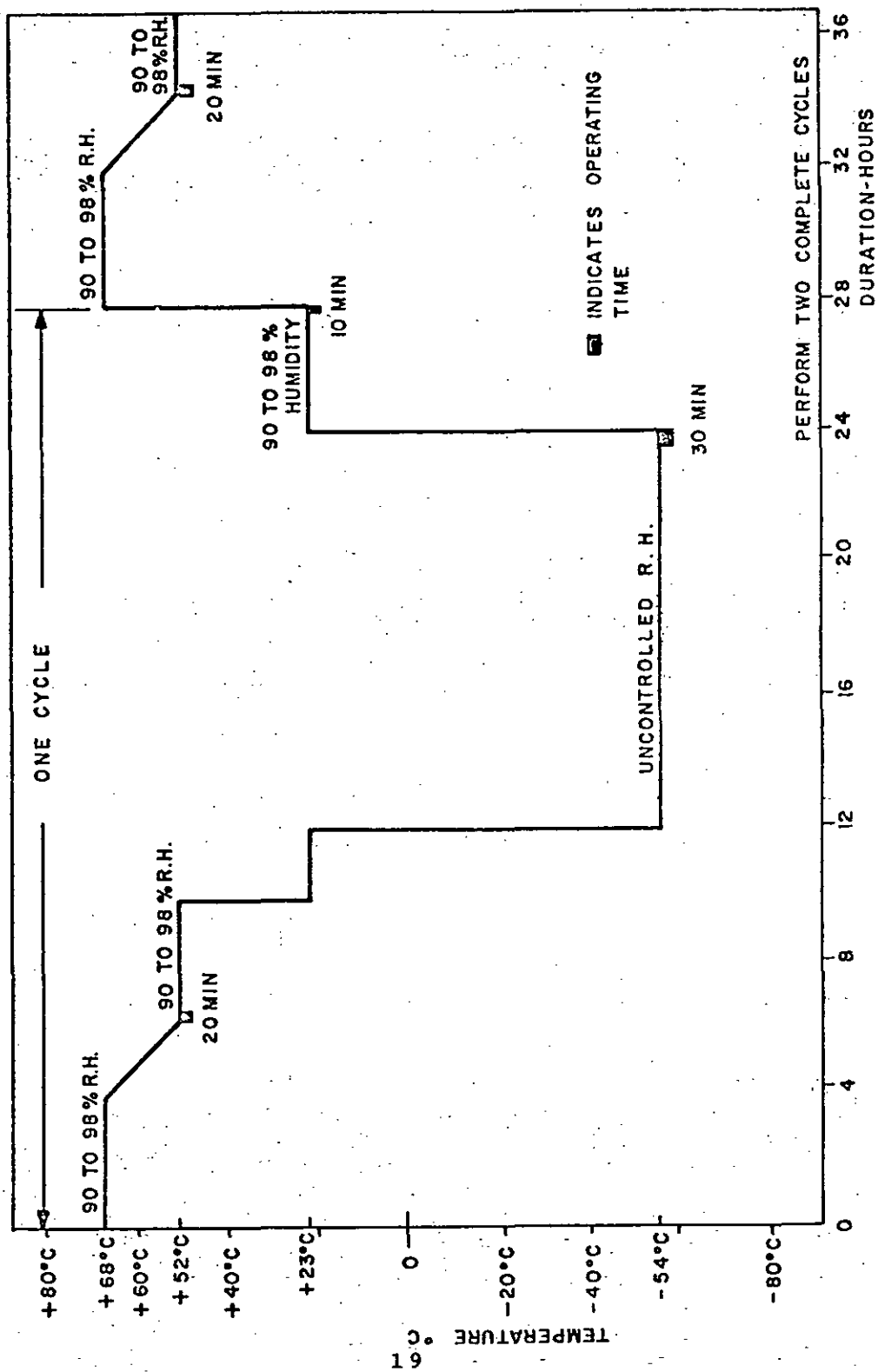


FIGURE 1. STORAGE, OPERATION AND THERMAL SHOCK

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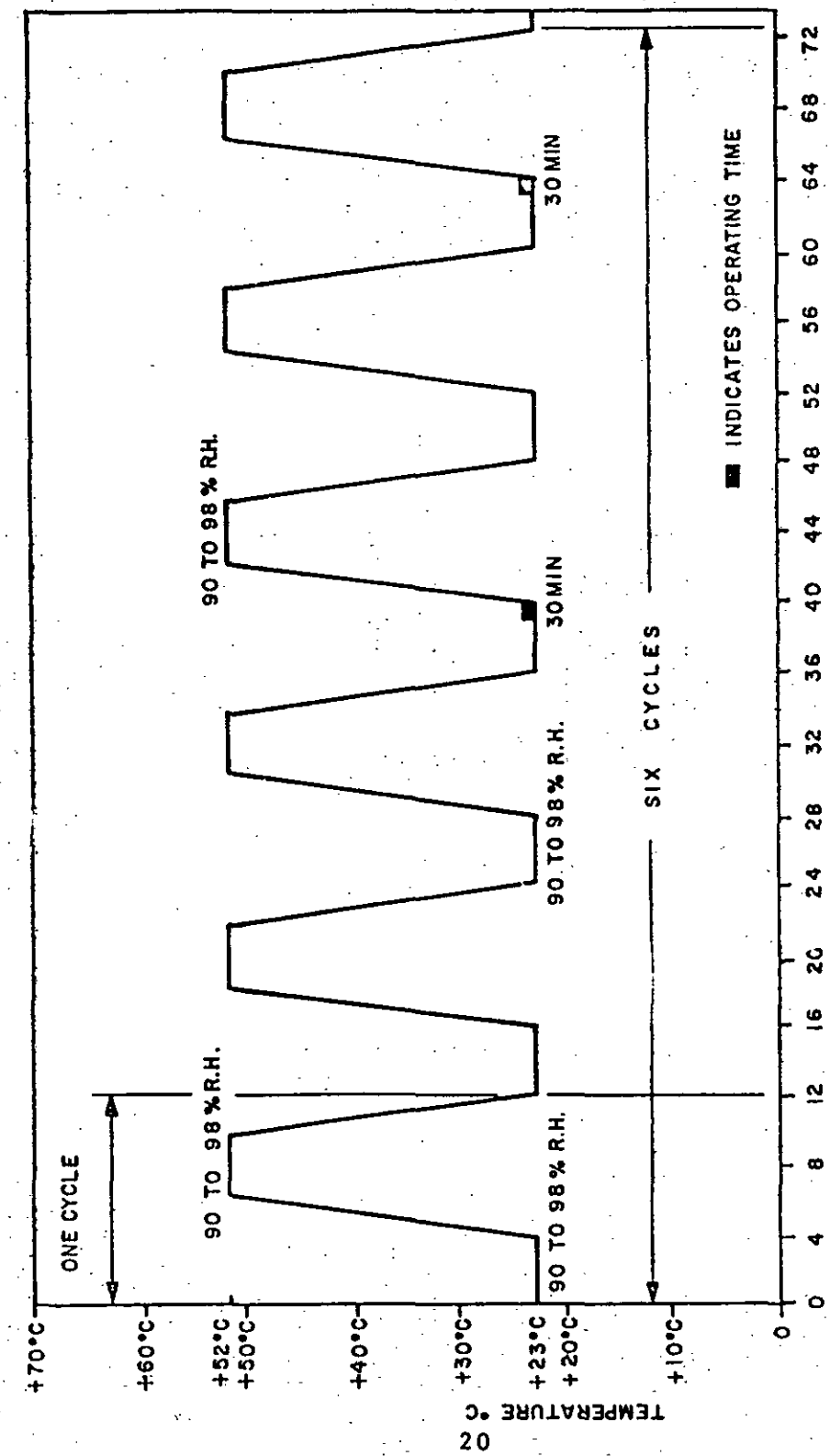
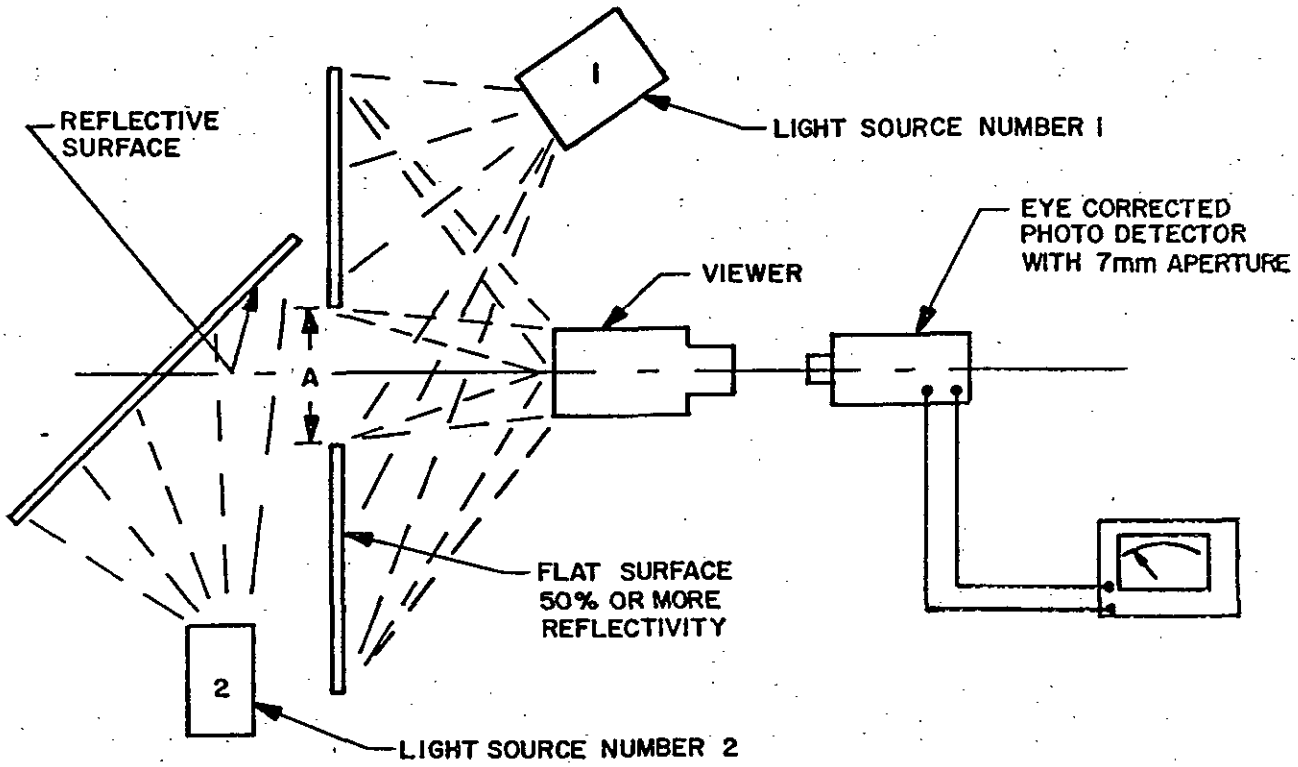


FIGURE 2. RELATIVE HUMIDITY TEST CYCLES

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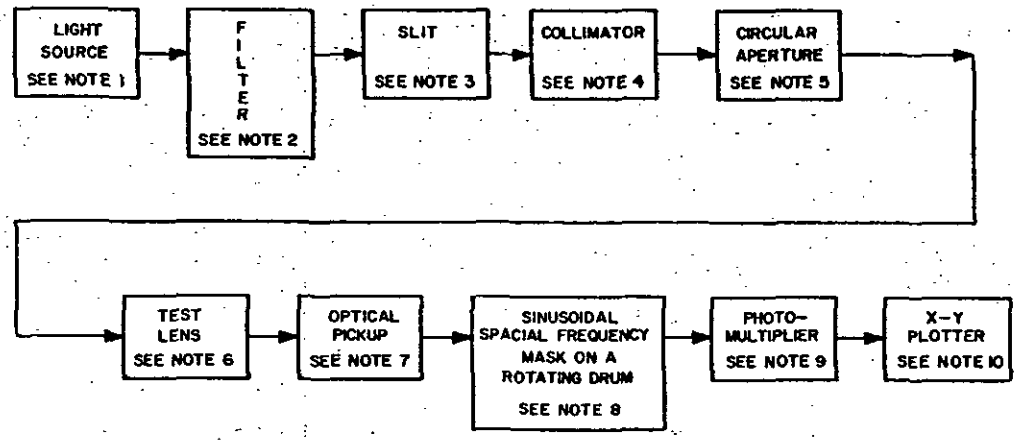


NOTES:

1. DIAMETER OF CIRCULAR HOLE "A" TO BE 6.0 INCHES LARGER THAN DIAMETER REQUIRED FOR ANGULAR SUBTENSE OF 6.5°, PLUS OR MINUS 0.5°.
2. SIZE OF REFLECTIVE SURFACE TO BE ADEQUATE TO FILL SOLID ANGLE RESULTING FROM DIAMETER NECESSARY UNDER NOTE NO.1 ABOVE
3. ANGULAR SUBTENSE OF FLAT SURFACE TO BE 120° MINIMUM.

FIGURE 3 STRAY LIGHT LUMINANCE

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Note 1 The light source shall be a tungsten D.C. filament ribbon lamp operating at a black body color temperature of 2870° K ± 100°K

Note 2 The filter shall conform to figure 7 to within 5.0%. The filter is used only when testing an eyepiece.

Note 3 The slit width shall be less than 2 microns in the image plane of the lens system in test. The slit height shall be at least ten times the slit width.

Note 4 The collimator shall have an EFL at least ten times the EFL of the lens system in test. The collimator shall have a clear aperture that completely fills the entrance aperture of the lens system in test.

Note 5 A circular aperture stop of 7 millimeters ± 0.1 millimeter shall be placed between the eyepiece and the collimator at the eye relief distance centered with respect to the optical axis of the eyepiece. The aperture stop shall not be used with objective measurements.

Note 6 The test lens shall be arbitrarily moved in relation to the rest of the MTF equipment in order to optimize the optical alignment. The parafocalized viewer (see note 7) shall be used to determine approximate optimumization and the x y plotter (see note 10) used to precisely determine optimization. Precise optimization shall be determined by obtaining maximum MTF at 30 ± 5 LP/mm on the output curve or both objectives and eyepieces. The eyepiece diopter adjustment may be arbitrarily adjusted between plus 1/2 to minus 1/2 diopters when attempting to optimize the optical alignment.

Note 7 The optical pickup shall be a 20 power 0.65 numerical aperture microscope objective used for measuring objectives and a 10 power 0.35 numerical aperture microscope objective used for measuring eyepieces. The pickup system shall contain a parafocalized eyepiece used for approximate focusing. Parafocalization is obtained between the sine wave mask and an eyepiece by adjusting the eyepiece focus.

Note 8 The sinusoidal mask shall have transmission continuously logarithmically increasing in frequency from 0.2 to 10.0 line pairs per millimeter with 2.0% distortion maximum and a density variation greater than 6 to 1.

Note 9 The photomultiplier shall have a S20 response conforming within the extremes of figure 8 with an anode dark current of 0.015 micro amps maximum.

Note 10 The X-Y plotter shall plot MTF on 2 cycle X 10 divisions per inch semi log paper 8 1/2 X 11 inches with 100 percent response equal to 5 inches on the the linear scale and spatial frequencies of 2 to 100 LP/mm on the log scale. The plotter shall comply with the following characteristics:

A. Frequency response	DC to 5 Hertz ± 3db
B. Flowing speed	20 in. per second
C. Repeatability	0.1% maximum variation
D. Accuracy	0.15% or less error
E. Sensitivity	100 millivolts per inch

Note 11 The overall system shall have an accuracy to within ± 3% and a repeatability to within ± 2%.

FIGURE 4 MODULATION TRANSFER FUNCTION TEST EQUIPMENT

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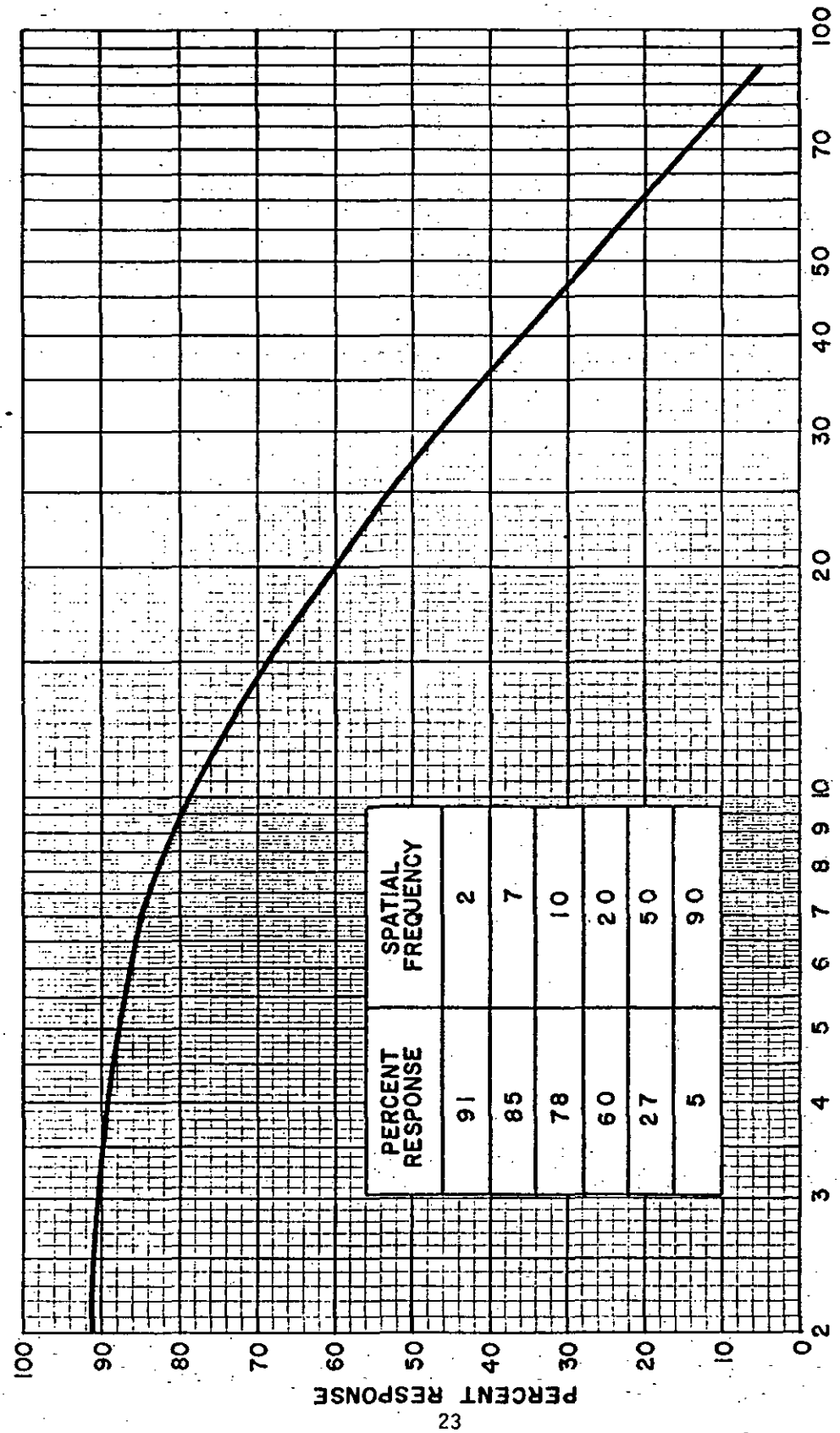


FIGURE 5. AN/TVS-2 OBJECTIVE MINIMUM RESPONSE

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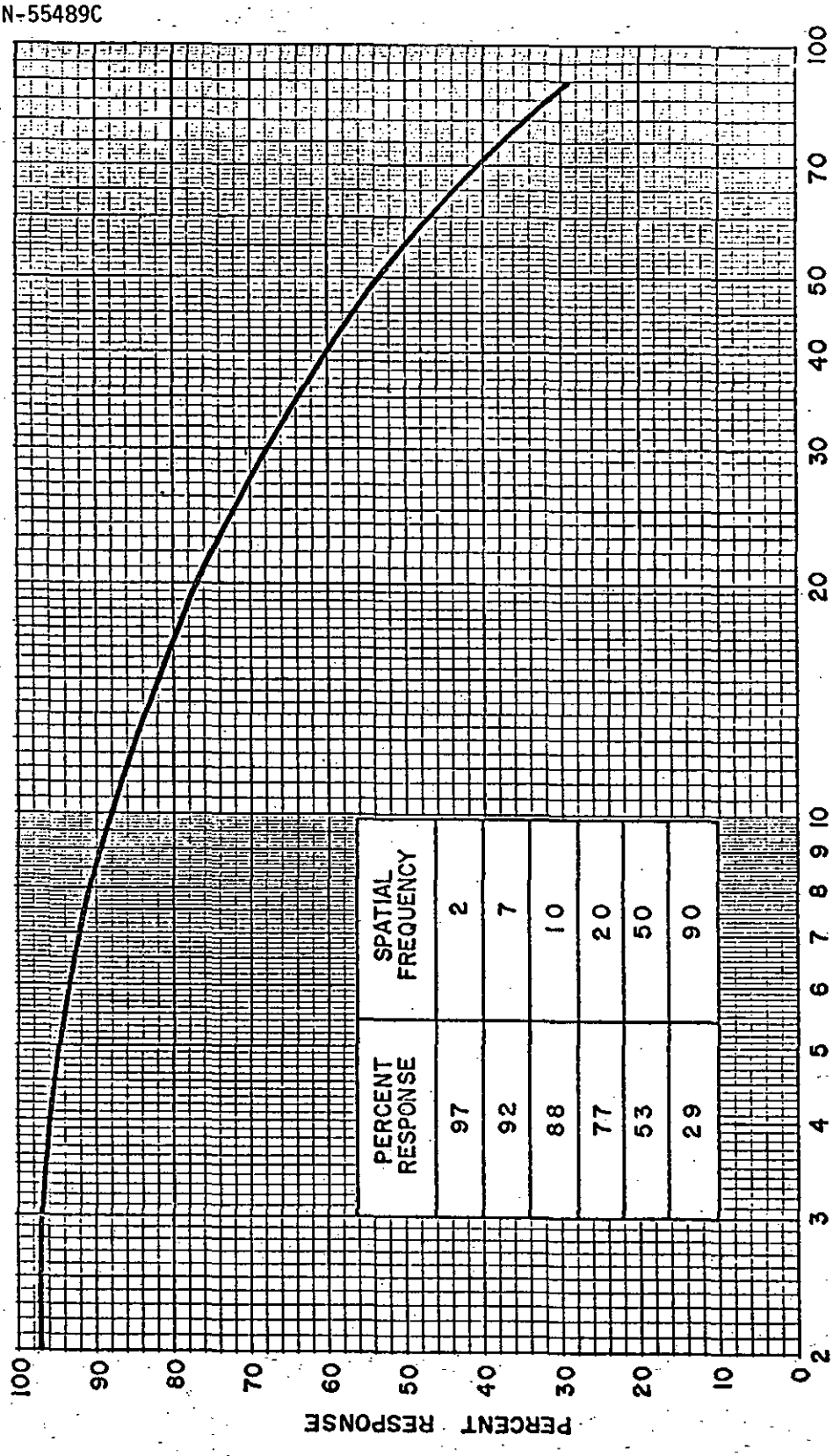


FIGURE 6 AN/TVS-2 EYEPIECE MINIMUM RESPONSE

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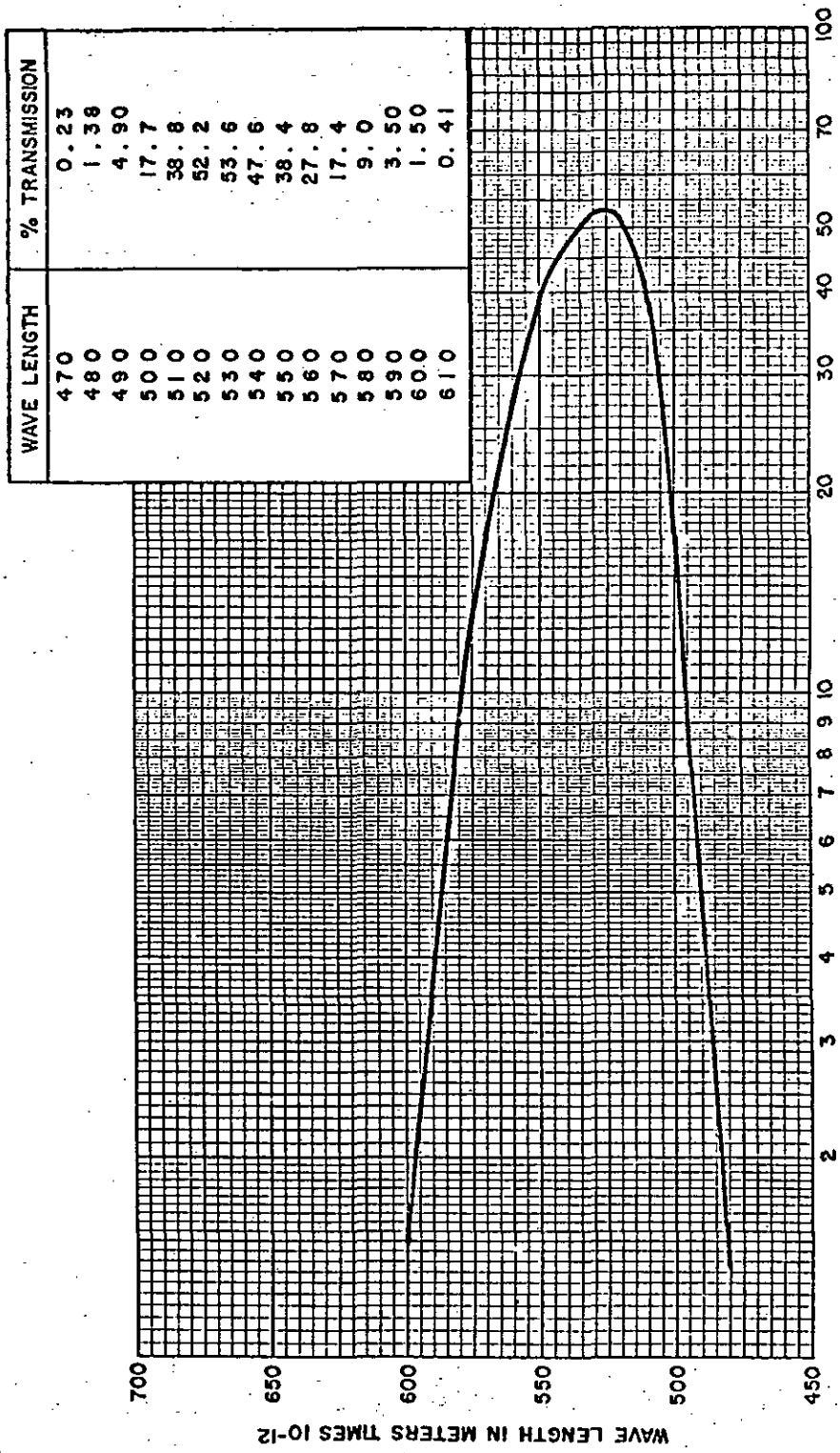


FIGURE 7 FILTER TRANSMISSION CHARACTERISTICS

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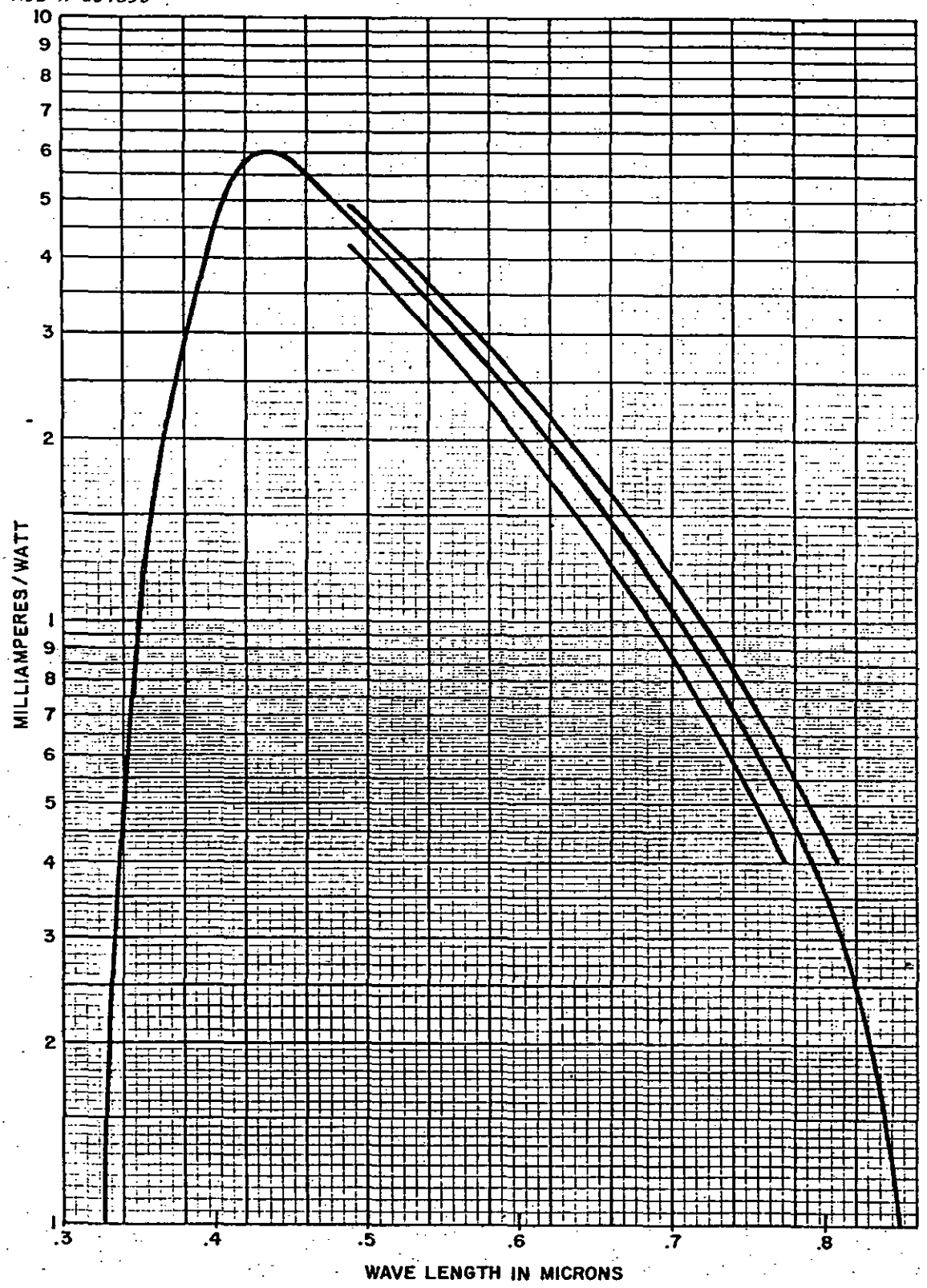


FIGURE 8 S20 SPECTRAL RESPONSE

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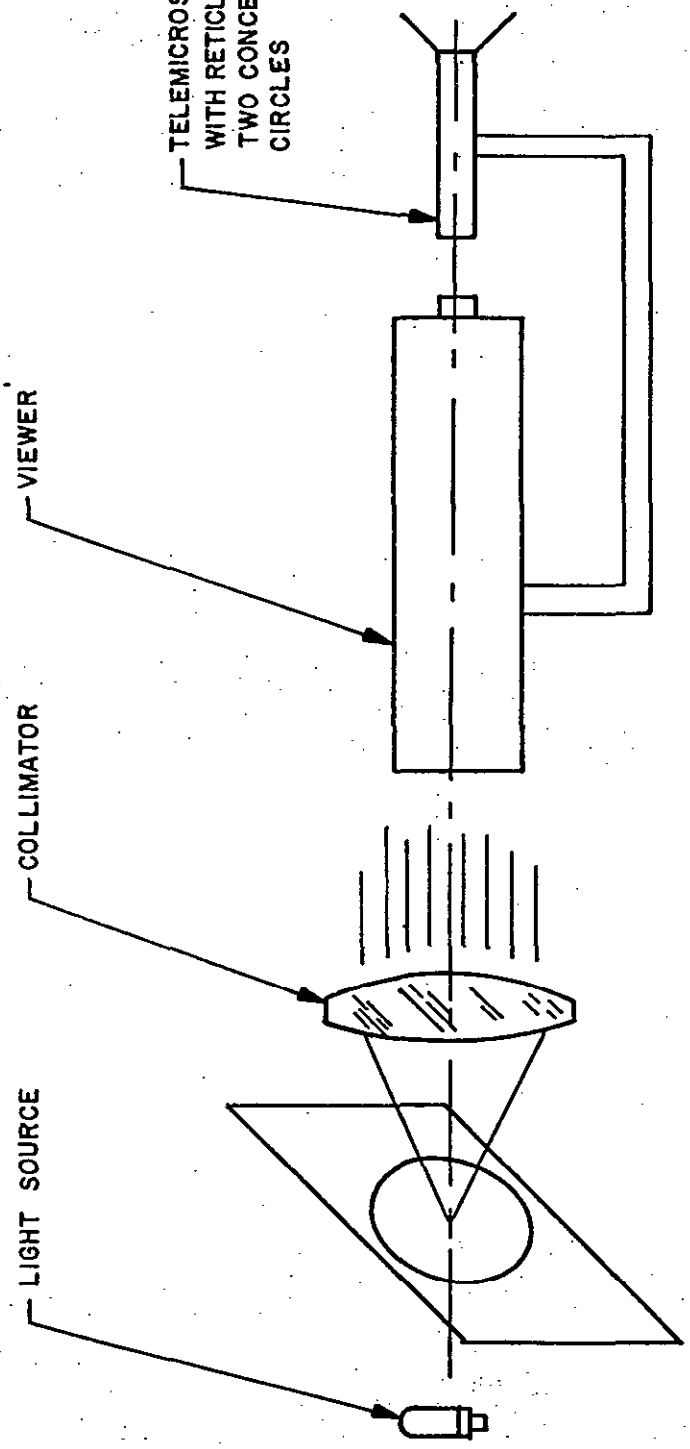


FIGURE 9. PERIPHERAL MAGNIFICATION

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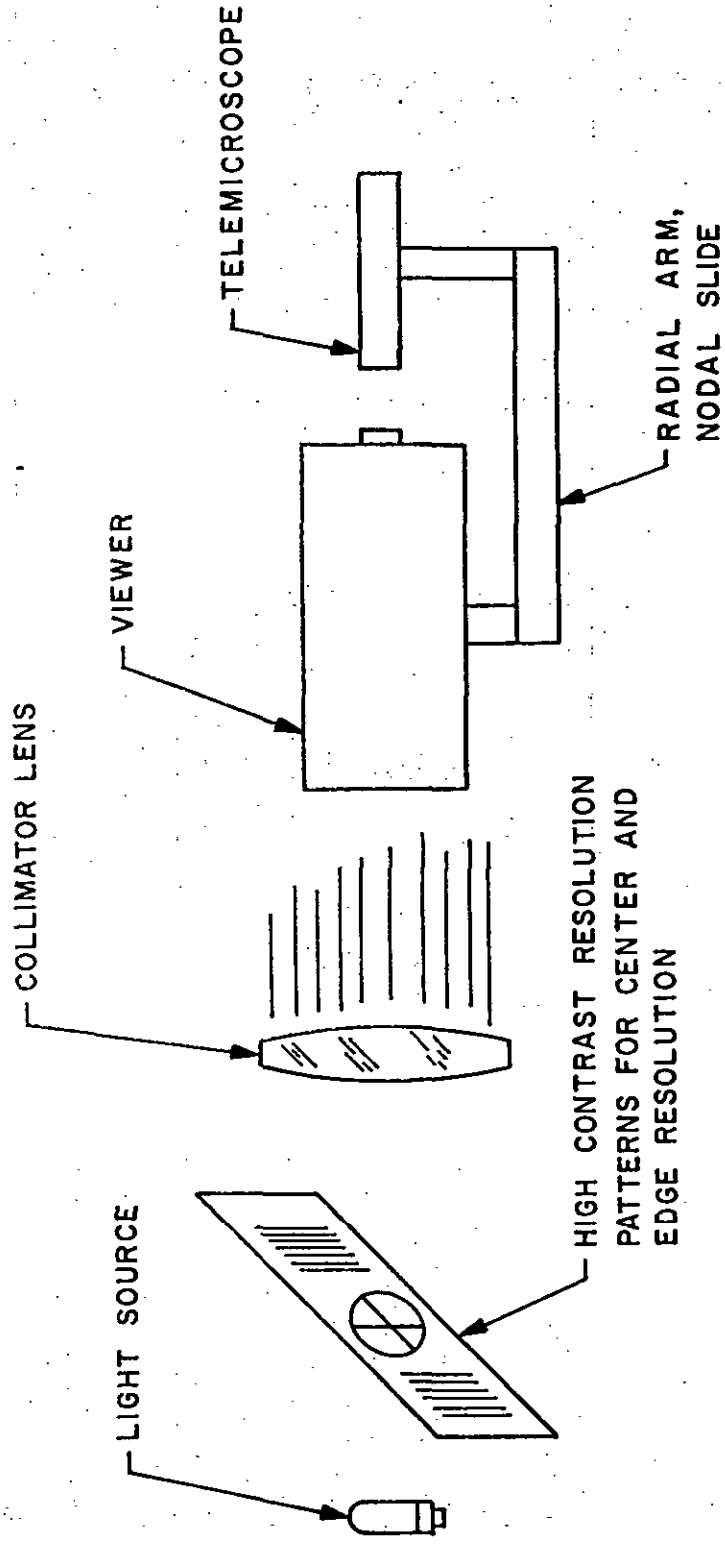
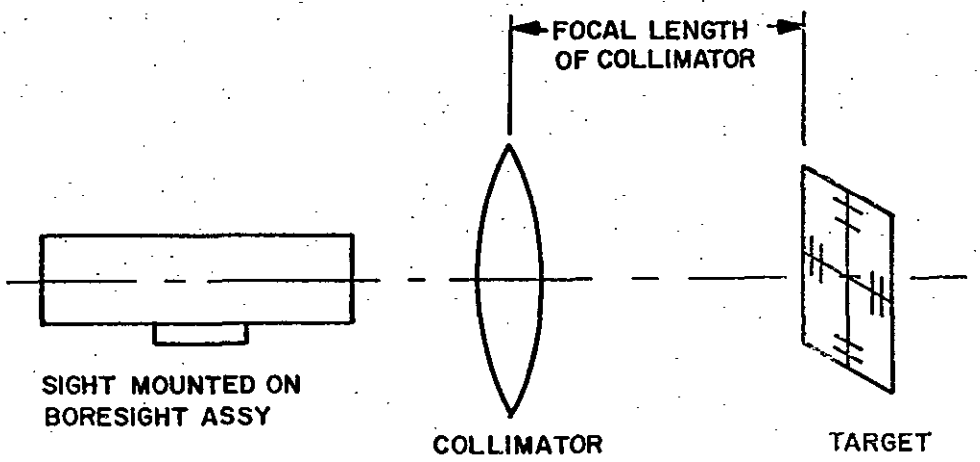
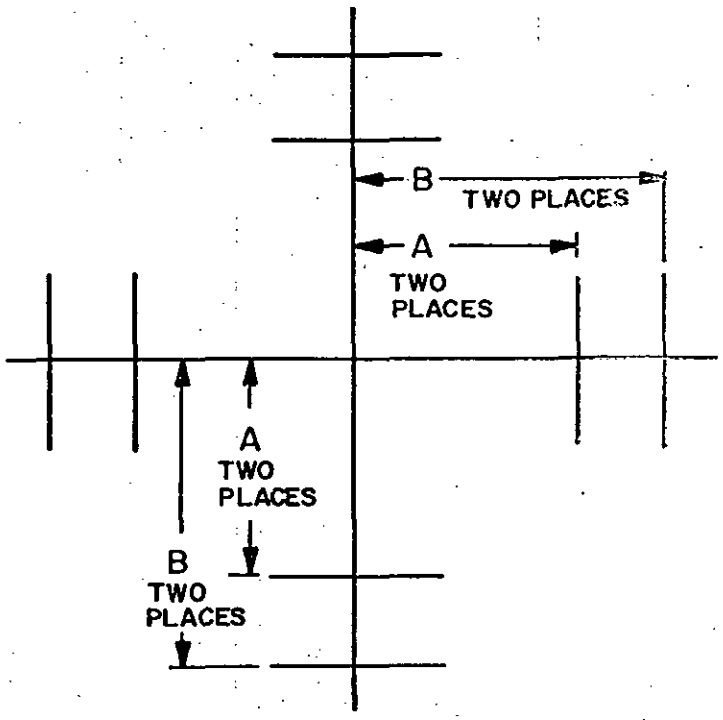


FIGURE 10

RESOLUTION, FIELD OF VIEW
AND RETICLE ADJUSTMENT TESTER

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A = FOCAL LENGTH IN INCHES OF COLLIMATOR X .00045 INCHES
B = FOCAL LENGTH IN INCHES OF COLLIMATOR X .00055 INCHES

FIGURE 11. CLICK MOVEMENT TEST PATTERN

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-N-55489C NIGHT VISION SIGHT CREW SERVED WEAPONS AN/TVS-2

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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